Lecture 2. Basic Quantities

Survival, Hazard, Cumulative Hazard, Quantiles, Residual Life

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1 Review: Classifying Incomplete Data

Situation	Type
Clinical trial starts at randomization (time 0); a participant completes 24 months with no event by study end.	RC
At baseline the tumour is already present; exact onset unknown, only that it occurred before baseline.	LC
Annual screening; disease-free at year 2, first positive at year 3.	IC
Registry enrols patients only if alive at clinic entry (some enter 3 years after diagnosis); followed until death or end of study.	LT + (RC or event)
Historical cohort assembled in 2020 from obituaries of people who died in 2010–2019; age at death recorded, no survivors.	RT

2 Survival Function

Let $X \geq 0$ be a random failure time.

Definition 2.1: Survival function

For continuous X with pdf f and cdf $F(x) = P(X \leq x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

$S(x)$ is the probability of surviving beyond time x. It is monotone non-increasing with $S(0) = 1$ and $S(\infty) = 0$.

Graphically, $S(x)$ is the area under the density to the right of x. (Typical examples are the Exp(0.5) density and the 1989 US all-cause survival curves by race/sex, which stay near 1 until middle age and then drop steeply.)

3 Hazard Function

Definition 2.2: Hazard function

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x \mid X \geq x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x)/P(X \geq x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx P(x \leq X < x + \mathrm{d}x) = S(x) - S(x + \mathrm{d}x)$ is the "infinitesimal" probability of failure at x, *unconditionally* on being alive before x.
- $h(x)\,\mathrm{d}x$ is the infinitesimal probability of failure in the next instant after x, *given* alive at x.

Remark 2.3: h is a rate, not a probability

$h(x) \geq 0$ can exceed 1; only $h(x)\Delta x$ for small Δx approximates a conditional probability.

3.1 Shapes of hazard functions

(i) **Constant**: e.g. the exponential distribution.

(ii) **Increasing**: industrial systems and cars, whose component failure rates rise over time (ageing).

(iii) **Decreasing** (less common): e.g. transplant studies (risk is highest just after the operation).

(iv) **Bathtub-shaped**: population mortality (high infant mortality, low in mid-life, rising in old age).

(v) **Hump-shaped (unimodal)**: rises early and eventually declines; e.g. breast cancer treatment studies where surgery- or toxicity-related high-risk patients tend to die early.

3.2 Examples

Example 2.4: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

The constant hazard is the *memoryless* property: $P(X > x + t \mid X > x) = e^{-\lambda t}$ does not depend on x.

Example 2.5: Weibull(α, λ)

$S(x) = \exp(-\lambda x^\alpha)$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^\alpha), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

$\alpha > 1$: increasing hazard; $\alpha = 1$: constant (exponential); $\alpha < 1$: decreasing.

4 Cumulative Hazard and the S–h Relationship

Motivation. Survival, density, hazard and cumulative hazard are four descriptions of the same distribution, and different methods estimate different ones. The relation $S(t) = \exp\{-H(t)\}$ lets us move between them.

Theorem 2.6: Continuous case

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If X is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

Proof [Supplement]. Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. Exponentiating gives $S(x) = e^{-H(x)}$. ∎

Summary of relations (continuous X)	
$f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$	$F(t) = 1 - S(t)$
$h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$	$H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$
$S(t) = \exp\{-H(t)\}$	$f(t) = h(t)\exp\{-H(t)\}$

Any one of f, F, S, h, H determines all the others.

5 Discrete Failure Times

At points $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum_{j=1}^{d} p_j = 1$. Discrete times arise from rounding of measurements or grouping failure times into intervals. Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

Definition 2.7: Discrete hazard

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^{d} p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability, so $0 < h(x_j) \leq 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

Example 2.8: $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2/3, & 1 \leq x < 2 \\ 1/3, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard equals 1: anyone alive just before the last support point must fail there.

Theorem 2.9: Product-limit representation

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})} \frac{S(x_{j-1})}{S(x_{j-2})} \cdots \frac{S(x_1)}{S(x_0)} S(x_0) = \prod_{i:x_i \leq x_j} \frac{S(x_i)}{S(x_{i-1})} = \prod_{i:x_i \leq x_j} [1 - h(x_i)],$$

using the telescoping product and $S(x_0) = S(0) = 1$.

This product form is the population version of the Kaplan–Meier estimator (Lecture 5).

Remark 2.10: Two definitions of the discrete cumulative hazard

Klein & Moeschberger define $H(x_j) = \sum_{i:x_i \leq x_j} h(x_i)$. With this definition $S(x) = \exp\{-H(x)\}$ does *not* hold. Cox & Oakes (1984) instead define

$$H(x_j) = -\sum_{i:x_i \leq x_j} \log\left[1 - h(x_i)\right] \approx \sum_{i:x_i \leq x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly (take logs of the product-limit form). The approximation uses $-\log(1-u) \approx u$ for small u.

6 Mean and Mean Residual Life

Motivation. For a patient who has survived to time t, the relevant question is the expected remaining lifetime. The mean residual life answers it, and it can be computed from the survival function alone.

Theorem 2.11: Mean as area under S

For $X \geq 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

Proof [Supplement]. Write $t = \int_0^t \mathrm{d}u$ and exchange the order of integration (Tonelli, integrand ≥ 0):

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty \int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

Definition 2.12: Mean residual life

$\mathrm{mrl}(x) = \mathrm{E}(X - x \mid X > x)$. For continuous X,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x) f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under the survival curve to the right of x divided by $S(x)$. Note $\mathrm{mrl}(0) = \mathrm{E}(X)$.

Proof of the second equality [Supplement]. Same Fubini argument with $t - x = \int_x^t \mathrm{d}u$: $\int_x^\infty (t-x) f(t)\,\mathrm{d}t = \int_x^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

7 Quantiles

Definition 2.13: p-th quantile

The p-th quantile ($100p$-th percentile) x_p satisfies $S(x_p) \leq 1 - p$:

$$x_p = \inf\{t : S(t) \leq 1 - p\}.$$

For continuous strictly decreasing S this is the unique solution of $S(x_p) = 1-p$, i.e. $F(x_p) = p$. The median is $x_{0.5}$.

Example 2.14: Exponential(λ) — mean, mrl, median

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{\int_x^\infty e^{-\lambda t}\,\mathrm{d}t}{e^{-\lambda x}} = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (constant: memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

Example 2.15: Weibull quantile

$S(x) = \exp(-\lambda x^\alpha)$, so $F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1 - p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \Big[\frac{-\ln(1-p)}{\lambda}\Big]^{1/\alpha}.$$

In particular the median is $x_{0.5} = (\ln 2/\lambda)^{1/\alpha}$.

Example 2.16: LogNormal quantile

$X \sim \text{LogNormal}(\mu, \sigma^2)$ means $Y = \ln X \sim N(\mu, \sigma^2)$. Since ln is strictly increasing, quantiles map through it:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

where Φ^{-1} is the standard normal quantile function. With $p = 1/2$, $\Phi^{-1}(1/2) = 0$, so $\text{median}(X) = e^\mu$.

8 Median Residual Life

Definition 2.17: Median residual life

Preferred to the mrl when the distribution is highly skewed.

$$\text{mdrl}(x) = \text{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \text{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \text{mdrl}(x),\ X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous X the numerator is $S(x + \text{mdrl}(x))$, hence

$$S\big(x + \text{mdrl}(x)\big) = \tfrac{1}{2}S(x) \quad \implies \quad \text{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

Example 2.18: Exponential(λ)

$S^{-1}(y) = -(1/\lambda)\ln y$ for $y \in (0, 1]$, so

$$\text{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}\big(-\lambda x - \ln 2\big) - x = \frac{\ln 2}{\lambda}.$$

Again constant in x.

9 Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^\alpha)$, $\lambda, \alpha > 0$, and $m(t) = \text{E}[T - t \mid T > t]$.

1. Derive $m(t)$.
2. Show $m(t)$ is constant when $\alpha = 1$.
3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

Derivation: substitution $v = \lambda u^\alpha$

With $f(u) = \lambda\alpha u^{\alpha-1}e^{-\lambda u^\alpha}$, the numerator is

$$N(t) = \int_t^\infty (u-t)\,\lambda\alpha u^{\alpha-1}e^{-\lambda u^\alpha}\,\mathrm{d}u.$$

Set $v = \lambda u^\alpha$, so $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$ and $u = (v/\lambda)^{1/\alpha}$; limits $u = t \mapsto v = \lambda t^\alpha$, $u \to \infty \mapsto v \to \infty$. Then

$$\begin{aligned} N(t) &= \int_{\lambda t^\alpha}^\infty \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t\Big)e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha}\int_{\lambda t^\alpha}^\infty v^{1/\alpha}e^{-v}\,\mathrm{d}v - t\int_{\lambda t^\alpha}^\infty e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1+\tfrac{1}{\alpha},\lambda t^\alpha\big) - t\,e^{-\lambda t^\alpha}, \end{aligned}$$

where $\Gamma(s,x) = \int_x^\infty v^{s-1}e^{-v}\,\mathrm{d}v$ is the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^\alpha}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)}\cdot\frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1+\tfrac{1}{\alpha},\lambda t^\alpha\big) - t.$$

- $\alpha = 1$: $\Gamma(2,\lambda t) = (1+\lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: the hazard $h(t) = \lambda\alpha t^{\alpha-1}$ is increasing, hence $m(t)$ decreases in t (proved in §10). Interpretation: the system *ages* — the longer it has survived, the shorter its expected remaining life.

10 mrl and Hazard Monotonicity

Motivation. Intuitively, if the risk of failure grows with age, the expected remaining life should shrink. This section makes that precise.

Theorem 2.19

Let $T \geq 0$ have survival S, density f, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if h is increasing then m is decreasing; if h is decreasing then m is increasing.

(1) Differential equation [Supplement]. $m(t) = \int_t^\infty S(u)\,\mathrm{d}u/S(t)$. By the quotient rule and $\frac{\mathrm{d}}{\mathrm{d}t}\int_t^\infty S(u)\,\mathrm{d}u = -S(t)$,

$$m'(t) = \frac{-S(t)\cdot S(t) - \big(\int_t^\infty S(u)\,\mathrm{d}u\big)S'(t)}{S(t)^2}.$$

Since $S'(t) = -f(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \big(\int_t^\infty S(u)\,\mathrm{d}u\big)h(t)S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

■

(2) Monotonicity [Supplement]. By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$ and $m'(t) \geq 0 \iff h(t)m(t) \geq 1$. Substituting $u \mapsto t+u$ and using $S(t+u)/S(t) = \exp\{-(H(t+u) - H(t))\}$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- If h is increasing, $h(s) \geq h(t)$ on $[t, t+u]$, so $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$ and

$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$

- If h is decreasing, $\int_t^{t+u} h(s)\,\mathrm{d}s \leq u\,h(t)$, so $m(t) \geq 1/h(t)$, $h(t)m(t) \geq 1$, and $m'(t) \geq 0$.

■

Example 2.20: Application to the Weibull

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: h decreasing $\Rightarrow$ m increasing.
- $\alpha = 1$: $h \equiv \lambda \Rightarrow m'(t) = \lambda m(t) - 1$; the bounded solution is $m \equiv 1/\lambda$ (so $m' = 0$).
- $\alpha > 1$: h increasing $\Rightarrow$ m decreasing.